



The effect of ammonia addition on the low-temperature autoignition of n-heptane: An experimental and modeling study

Liang Yu, Wei Zhou, Yuan Feng, Wenyu Wang, Jizhen Zhu, Yong Qian, Xingcai Lu*

School of Mechanical Engineering, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai 200240, PR China

ARTICLE INFO

Article history:

Received 2 January 2020

Revised 13 March 2020

Accepted 17 March 2020

Keywords:

N-heptane/ NH_3

Ignition delay time

NH_3 fraction

Blending mechanism

Rapid compression machine

ABSTRACT

Ammonia (NH_3) is receiving increasing attention as an alternative engine fuel due to its carbon-free nature. However, its fundamental combustion characteristics, such as higher autoignition temperature and lower burning velocity compared to conventional hydrocarbons, limit its direct use in traditional engines. To confront this dilemma, hydrocarbon/ NH_3 blending fuels are seen as one of the most appropriate ways to exploit its advantage and offset its weaknesses. Using n-heptane as a representative of hydrocarbons, this study investigated the effect of NH_3 addition on the low-temperature autoignition of n-heptane. The ignition delay times of five n-heptane/ NH_3 mixtures with NH_3 fractions of 0%, 20%, and 40% were measured in a rapid compression machine at temperatures of 635–945 K, pressures of 10 and 15 bar, and equivalence ratios of 1.0 and 2.0. Experimental results show that the n-heptane/ NH_3 blending fuels exhibit pronounced low-temperature reactivity, and both the total and the first-stage ignition delay times increase with the increase of NH_3 fraction. A blending mechanism of n-heptane/ NH_3 was compiled based on the existing n-heptane mechanism and NH_3 mechanism. It is found that the blending mechanism is capable to predict the inhibition effect of NH_3 addition on n-heptane autoignition and qualitatively capture the dependence of ignition delay time on equivalence ratio and oxygen mole fraction. Nevertheless, there are significant discrepancies between the experiments and the simulation. Furthermore, kinetic analyses, including species evolution, rate of production and sensitivity of OH radical were conducted sequentially to reveal the autoignition kinetics of NH_3 /n-heptane blends and the interaction between n-heptane and NH_3 . Suggestions are provided for the further development of the blending mechanism.

© 2020 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Ammonia has long been recognized as an effective agent for selective non-catalytic reduction (SNCR) [1], which has been widely used in the NO_x control of modern engines. In recent years, ammonia has once again attracted extensive attention as a carbon-free and energy-carrying fuel. Compared to hydrogen, NH_3 has obvious advantages of production, storage, and transportation [2]. These features make ammonia a potential and desirable alternative engine fuel. Therefore, it is necessary for the community to understand the characteristics and mechanisms of ammonia combustion at engine relevant conditions, of which its autoignition behavior is vitally important.

Experimental studies on NH_3 autoignition can be divided into two stages. In the 1960–1980s, shock tube (ST) investigations

[3–10] were carried out to reveal the oxidation kinetics of NH_3 at temperatures higher than 1500 K. Those earlier studies provide basic knowledge for NH_3 oxidation mechanism. In the last 5 years, NH_3 has gained increasing attention as a promising fuel. More experiments were performed in STs [11,12] and rapid compression machines (RCMs) [13,14] to report the ignition delay time (IDT) of NH_3 and NH_3/H_2 blend. Mathieu and Petersen [12] measured the IDTs of NH_3 in highly diluted Ar in a shock tube at temperatures of 1560–2455 K, pressures of 1.4, 11 and 30 atm, and equivalence ratios of 0.5, 1.0 and 2.0. An improved NH_3 kinetic model was developed and successfully used to model the measured IDTs. Shu et al. [11] reported the IDTs of NH_3 /air mixture measured in a shock tube at the temperature range of 1100–1600 K, two pressures of 20 and 40 bar, and equivalence ratios of 0.5, 1.0 and 2.0. It is shown that the experimental results agree well with the predictions of Mathieu and Petersen's mechanism [12]. Pochet et al. [13] first used RCM to investigate NH_3 autoignition. The IDTs of NH_3 with hydrogen addition (0%, 10%, and 25%) were measured in a narrow temperature range of 1000–1100 K, pressures of 43 and 65 bar,

* Corresponding author.

E-mail address: lyuxc@sjtu.edu.cn (X. Lu).

Table 1

A summary of the modeling studies on ammonia combustion.

Model	Component	Species/Rxn	Validation
Miller et al. (1983) [16]	NH ₃	22/98	Flame speed and speciation
Lindstedt et al. (1994) [17]	NH ₃	22/95	Flame speciation and temp.
Tian et al. (2009) [18]	NH ₃ /CH ₄	84/703	Flame speciation and temp.
Mendiara and Glarborg (2009) [19]	NH ₃ /CH ₄	97/779	Flow reactor speciation
Mathieu and Petersen (2015) [12]	NH ₃ /CH ₄	54/278	Ignition delay time (IDT)
Nakamura et al. (2017) [20]	NH ₃	38/232	Flame speciation
Glarborg et al. (2018) [15]	NH ₃ /C ₁ –C ₂	151/1397	Speciation and IDT
Shrestha et al. (2018) [21]	NH ₃	125/1090	LBVs, speciation and IDT
Mei et al. (2019) [22]	NH ₃	37/255	LBVs

and equivalence ratios of 0.2–0.5. Recently, He et al. [14] studied the autoignition of NH₃/H₂ mixture (1–20% mol. H₂) in an RCM for temperatures ranging from 950 to 1150 K, pressures of 20–60 bar, and equivalence ratios of 0.5–2.0. Both studies find that H₂ addition significantly promotes NH₃ autoignition. In sync with the experimental research, there is also a lot of modeling work on NH₃ combustion as listed in Table 1. Those models have been proven to successfully predict the high-temperature performance of NH₃ and NH₃/CH₄, including ignition delay time, flame speed and speciation characteristics. It is worth mentioning that the recent work of Glarborg et al. [15] presents the complete nitrogen chemistry including NH₃ and C₁–C₂ hydrocarbons sub-mechanisms with the latest quantum calculation results of some key reactions.

It is understandable that the existing studies mainly centered on the high-temperature (above 1000 K) performance of NH₃ oxidation and combustion because NH₃ does not show low-temperature (low-T) chemistry as n-heptane. Besides, recent researches confirm that the laminar burning velocity (LBV) of NH₃ is much lower than hydrocarbon fuels [23,24]. The two disadvantages (higher autoignition temperature and lower burning velocity) limit the direct use of NH₃ in compression ignition (CI) and spark ignition (SI) engines. At this stage, one of the most feasible ways to use NH₃ in engine combustion is to blend it with conventional hydrocarbons whether from a technical or economic perspective, such as ammonia-diesel blending fuel [25]. Since engine combustion typically covers a wide range of temperature conditions, the low-T combustion characteristic of NH₃/hydrocarbon fuels does matter and cannot be ignored. However, experimental and modeling studies on the fundamental combustion behavior of NH₃/hydrocarbon (larger than CH₄) blends are still lacking, especially in the low-T range of 600–1000 K, which greatly confines the combustion/fuel design of engines feeding with NH₃. In this study, n-heptane, the simplest diesel surrogate, was chosen as a representative of hydrocarbon fuels. The low-T autoignition behaviors of n-heptane/NH₃ blends were investigated at wide conditions in a rapid compression machine. The effect of NH₃ addition on the low-T ignition delay time of n-heptane was experimentally revealed. At the same time, a blending mechanism was constructed and validated against the experimental results. Overall, this study provides the n-heptane/NH₃ blend IDT data which will be useful for developing blend mechanism.

2. Methodology

2.1. Rapid compression machine

The experiments of n-heptane/NH₃ blends were carried out in SJTU-RCM [26–28]. Here is a brief introduction to the device. The fundamental principle of the RCM is an adiabatic compression process to create a controllable pressure (*P*) and temperature (*T*) environment, similar to a single compression stroke of diesel engines. The main body of the RCM is divided into three parts, the driver part, the braking part, and the combustion part. The driver part

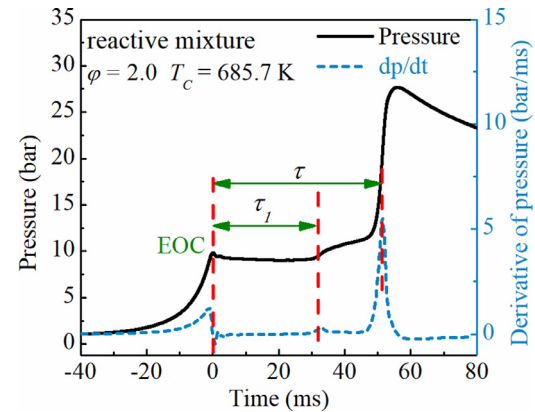


Fig. 1. The autoignition pressure curve of n-heptane/NH₃ blend (mixture 4) at the condition of $\phi = 2.0$, $T_c = 685.7$ K, $P_c = 10$ bar.

provides driving force by the pressure difference between the front and rear of a 125 mm driver piston. The function of the braking part is to brake the moving system at the end of compression (EOC). The combustion part is the core region of the RCM where fuel autoignition happens and is detected. It is equipped with a well-designed preheating and temperature control system, so the initial temperature (T_0) of the reactants can be precisely designated. The diameter of the combustion chamber and the creviced piston is 35 mm. The RCM has a fixed stroke of 215 mm and a variable compression ratio (CR) between 4 and 20. Initial pressure (P_0) is detected by a digital pressure gate (with a minimum scale of 0.01 kPa). Dynamic pressure inside the combustion chamber before and after compression is measured by a piezoelectric transducer (Kistler 6125B). The pressure and temperature at EOC are referred to as compressed pressure (P_c) and compressed temperature (T_c). The compressed temperature, T_c , cannot be directly measured and is determined by “adiabatic core” hypothesis [29], $\int_{T_0}^{T_c} \frac{\gamma(T)}{\gamma(T)-1} \frac{dT}{T} = \ln[P_c/P_0]$. In this equation, $\gamma(T)$ refers to the specific heat ratio of reactant mixture, which is dependent on temperature. In this study, T_c was gradually elevated in a step of 20–30 K by increasing CR or T_0 . The systematic uncertainty of IDTs varies with many factors [30]. In this study, the uncertainty of the measured IDT is less than $\pm 15\%$, of which the uncertainty analysis method has been specified in our earlier work [28].

Figure 1 is a typical pressure curve (solid line) of n-heptane/NH₃ blend autoignition in this RCM. The total ignition delay time (τ) refers to the time duration between EOC and the maximum dp/dt point, as shown in Fig. 1. Similarly, the first-stage ignition delay time (τ_1) is defined as the time from EOC to the first local dp/dt maximum point. First-stage ignition is a sign of low-T reactivity which is one of the focuses of this paper. There is no doubt that NH₃ has a great impact on the first-stage ignition of

Table 2
Reactant mixtures used in the experiment.

Mixture	φ	Ar/O ₂	Share of energy		Mole fraction		
			NH ₃	C ₇ H ₁₆	NH ₃	C ₇ H ₁₆	O ₂
Mixture 1	1.0	8.52	0%	100%	0.00%	0.95%	10.40%
Mixture 2	1.0	8.52	20%	80%	2.64%	0.74%	10.15%
Mixture 3	1.0	8.52	40%	60%	5.18%	0.55%	9.90%
Mixture 4	2.0	8.52	40%	60%	9.80%	1.03%	9.37%
Mixture 5	1.0	3.76	40%	60%	9.80%	1.03%	18.73%

n-heptane/NH₃ blends, though NH₃ itself does not exhibit low-T reactivity.

2.2. Reactant mixture

As noted above, blending NH₃ with conventional hydrocarbon fuels may be the most viable method of burning NH₃ in engines. However, most of related fundamental combustion researches were still limited to the performance of the mixture of NH₃ and C₀–C₁ hydrocarbons which are far from the composition of actual gasoline, kerosene, and diesel fuels. In this study, n-heptane was chosen to represent hydrocarbons for three reasons. It is the most commonly used surrogate component in real fuel surrogates. It owns the typical characteristics of actual fuels, namely low-T reactivity. Most importantly, it has the most mature reaction kinetic model, which is important for the assessment of the amine subset of hydrocarbon/NH₃ blending mechanism.

We adopted share of total energy, a common concept in engine experiment, to characterize the fraction of NH₃ in the n-heptane/NH₃ blends. For example, a NH₃ content of 20% means that NH₃ accounts for 20% of the total energy of the mixing fuel. In this study, five reactant mixtures with NH₃ blending ratios of 0%, 20%, and 40% were investigated, as shown in Table 2. Ar instead of N₂ was used as the buffer gas to lower the preheating temperature. With the low heat values (LHVs) of n-heptane (4500.282 kJ/mol) and NH₃ (316.8675 kJ/mol), the energy fractions are easily converted into mole fractions. To facilitate the direct use of the mixture composition, Table 2 also lists the mole fractions of the reactants. Because ammonia easily sticks to the mixing tank walls, pure NH₃ was fed into the tank to saturate it before preparation the reactive mixture. It should be noted that the heat capacity of the mixture increases with the increase of NH₃ fraction. The consequent result is that to reach the same compressed temperature, the higher NH₃ content mixture needs a higher compression ratio. Besides, for the same heat release or heat loss, the mixtures with higher heat capacity will have larger temperature variation. Considering that the difference of different mixtures in heat capacity is small (less than 5%), it will not have an obvious effect on the ignition delay time.

2.3. Kinetic mechanism

The kinetic models of n-heptane and NH₃ are already available in literature. However, even with mature n-heptane and NH₃ sub-mechanisms, developing a complete n-heptane/NH₃ blending mechanism is not a one-step task. This is because complete hydrocarbon/NH₃ blending mechanisms must include the interaction between nitrogen and carbon chemistry, but research on this aspect are still very scarce. In this study, we compiled an n-heptane/NH₃ blending mechanism to predict the n-heptane/NH₃ blends autoignition. The n-heptane mechanism is from Zhang et al. [31] which has been widely validated and is found to well capture the newly measured IDTs of n-heptane, as shown in Fig. 2. The NH₃ mechanism is from Glarborg et al. [15], including the latest theoretical calculation achievements and NH₃/C₁–C₂ chemistry,

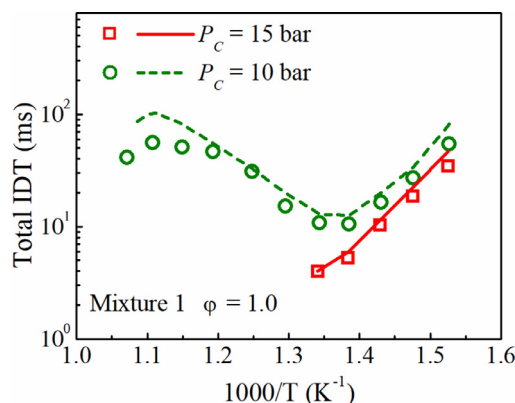


Fig. 2. Total IDTs of mixture 1 (neat n-heptane) at an equivalence ratio of 1.0 for pressures of 10 and 15 bar. Symbols are the experiment results of this study and lines are the simulation results using Zhang's n-heptane mechanism [31].

which has been validated against high-temperature oxidation and autoignition [15]. Therefore, the blending mechanism may be the optimal one based on current research achievement. During the merger process, the NH₃ sub-mechanism (excluding H₂/C₀–C₁ core mechanism) was add to the complete n-heptane mechanism. The same is true for thermodynamics. CHEMKIN PRO software [32] is adopted for the simulation by combining constant volume reactor and variable volume profiles deduced for non-reactive experiments [28]. The variable volume profiles of the reported IDT data are shown in the Supplementary material.

3. Experimental and simulation results

One of the main purposes of this study is to reveal the effect of NH₃ addition on the low-T autoignition of n-heptane. Figure 3 shows the comparison of the total and first-stage IDTs of the n-heptane/NH₃ blends with NH₃ fractions of 0%, 20%, and 40% at the equivalence ratio of 1.0 for two pressures of 10 and 15 bar. As expected, the IDTs get longer with increasing NH₃ blending fraction. At the same time, the total IDT of the three blends exhibit similar NTC behavior whose intensity and temperature range do not change as the change of NH₃ fraction. The first-stage IDTs behave Arrhenius-type relation with temperature at the investigated conditions. The first-stage IDTs of the three blends are almost parallel, indicating that they have a similar temperature dependence. When the temperature is higher than 720 K, entering the NTC region, the difference in the total IDT of the three blends is expanded. Figure 4 contrasts the experimental autoignition pressure curves of the three blends to further explore the NH₃ addition effect. Apart from the numeral difference of IDT, it is worth noting that the ratio of the first-stage IDT in the total IDT (τ_1/τ) decreases with the increase in NH₃ fraction in fuel blend, indicating that NH₃ has a greater impact on the second-stage IDT ($\tau - \tau_1$) than the first-stage IDT.

In terms of simulation, it is shown in Fig. 3 that the blending mechanism can capture the total and first-stage IDTs of mixture 1, especially in the NTC and lower temperatures, which is a reflection of the well-established n-heptane sub-mechanism [31] and is independent of NH₃ subset. For mixtures 2 and 3, although the blending mechanism can predict the inhibition effect of NH₃ addition on n-heptane autoignition, it fails to quantitatively capture the IDTs over the whole investigated temperatures. At temperatures below 720 K, the simulated total IDTs are much higher than those of the experiment, and the influence of NH₃ addition on n-heptane autoignition is greatly overpredicted. At the pressure of 10 bar, the blending mechanism predicts close total IDTs between mixture 2 and mixture 3 in the NTC region, which is contrary to the exper-

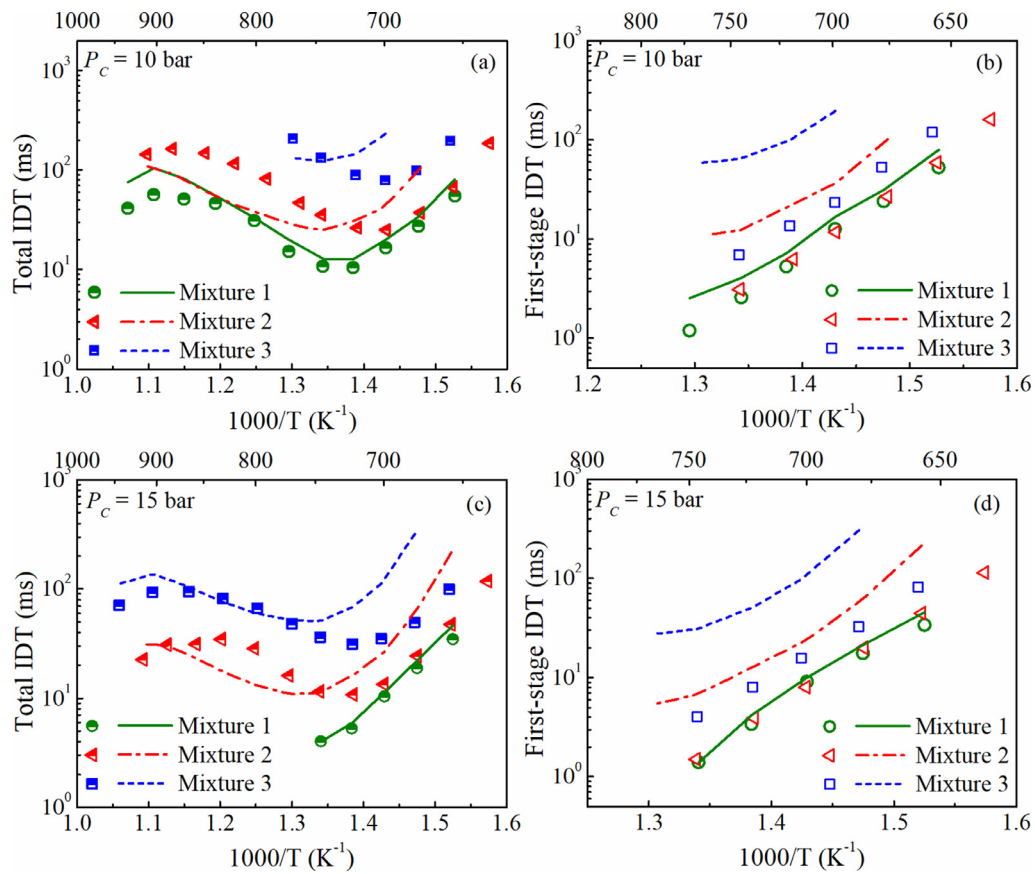


Fig. 3. Comparison of the IDTs of n-heptane/NH₃ blending fuels with NH₃ fractions of 0% (mixture 1), 20% (mixture 2), and 40% (mixture 3). Dots are the experimental results and lines are the simulation results.

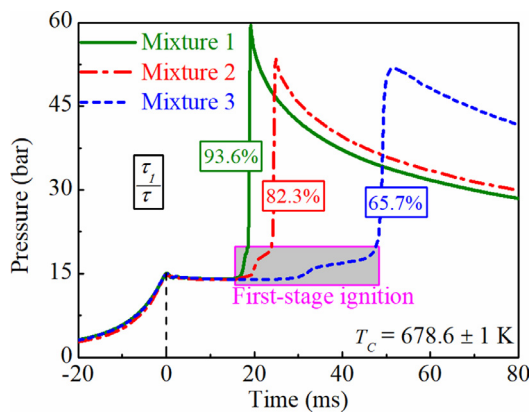


Fig. 4. Comparison of the experimental autoignition pressure curves of n-heptane/NH₃ blending fuels with NH₃ fractions of 0% (mixture 1), 20% (mixture 2), and 40% (mixture 3).

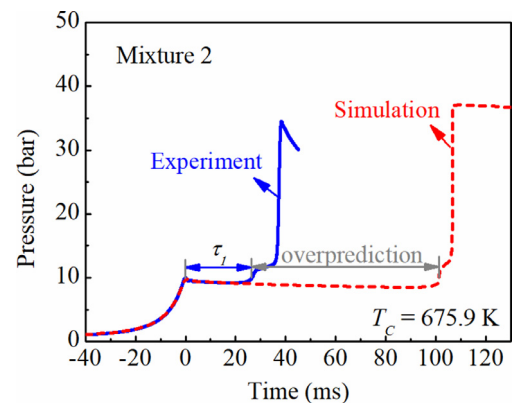


Fig. 5. Comparison of the autoignition pressure curves of mixture 2 between the experiment and the simulation.

imental measurements. In addition, the blending mechanism also greatly overestimates the first-stage IDT of the two blends while it captures the IDT slope, as shown on the right of Fig. 3. Another noteworthy finding is that the mechanism overestimates the NTC initiation temperature of the two blends, which is believed to be caused by deficiencies of the low-T subset of the blending mechanism.

Figure 5 compares the experimental and simulated autoignition pressure curves of mixture 2. The figure reveals that the big differences in total IDT between the experiment and the simulation are mainly because that the mechanism excessively overesti-

mates the first-stage IDT at the investigated condition. The low-T performance of the blending mechanism depends on three factors: the n-heptane sub-mechanism, the NH₃ sub-mechanism, and the interaction between the two. The present state is that the low-T adaptability of the NH₃ oxidation sub-mechanism has not been experimentally verified, and the cross reactions between nitrogen and carbon chemistry (C₂–C₇) are still missing. On the contrary, n-heptane sub-mechanism has already been well understood and widely validated. Therefore, we have good reasons to speculate that the prediction deviations of the blending mechanism may mainly come from the latter two factors. There is still a long way to complete the blending mechanism.

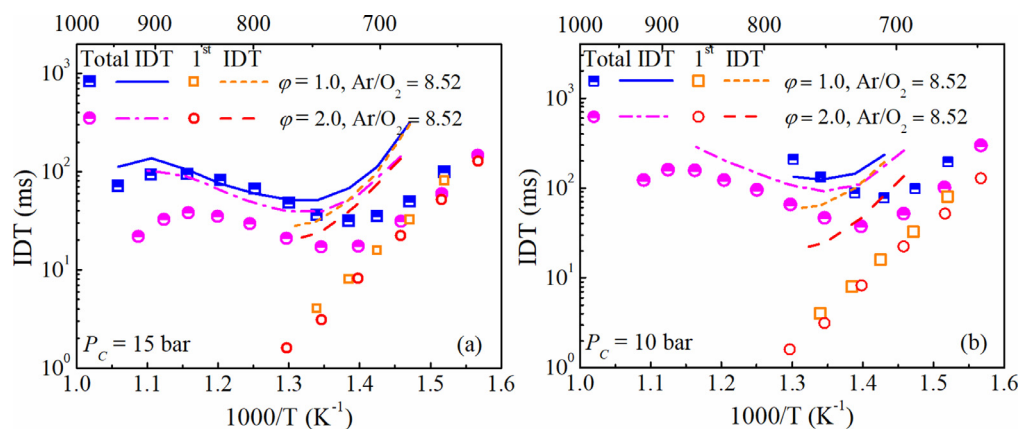


Fig. 6. Comparison of IDTs of 40% NH₃ blends at different equivalence ratios of 1.0 (mixture 3) and 2.0 (mixture 4).

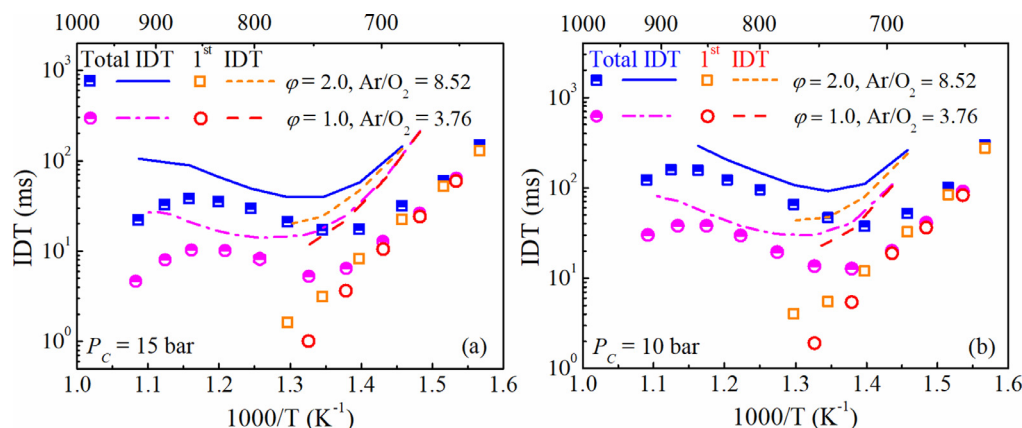


Fig. 7. Comparison of IDTs of 40% NH₃ blends at different oxygen mole contents of 9.37% (mixture 4) and 18.73% (mixture 5).

Additionally, experimental and modeling studies were also conducted for another two 40% NH₃ blends, mixtures 4 and 5. This section aims to reveal the dependence of IDTs on equivalence ratio and oxygen mole fraction, the two important parameters in engine combustion. It is shown in Figs. 6 and 7 that both equivalence ratio and oxygen mole fraction show a negative relation with the IDTs of the blending fuels, which is similar to the autoignition characteristics of hydrocarbons investigated in our earlier work [26–28]. The blending mechanism fails to quantitatively predict the IDTs but is capable of qualitatively predicting the effect of equivalence ratio and oxygen mole content on IDT. In addition, it is shown in Fig. 6 that the measured first-stage IDT only slightly increases (even overlaps) as the equivalence ratio rising from 1.0 to 2.0, while the simulation result triples at most. This finding indicates that the equivalence ratio dependence of IDT is overpredicted by the blending mechanism.

4. Kinetic analysis

To reveal the autoignition kinetics of NH₃/n-heptane blends and the interaction between n-heptane and NH₃, kinetic analyses including species evolution, rate of production (ROP) and sensitivity analysis of OH radical were performed sequentially based on the blending mechanism under constant volume simulation. Figure 8 takes mixture 2 as a representative to present reactant consumption and OH radical evolution profiles during the autoignition. It is shown that n-heptane and NH₃ are not consumed simultaneously. The whole reactivity of the blending fuel is driven by n-heptane, which is featured by the obvious n-heptane consumption before

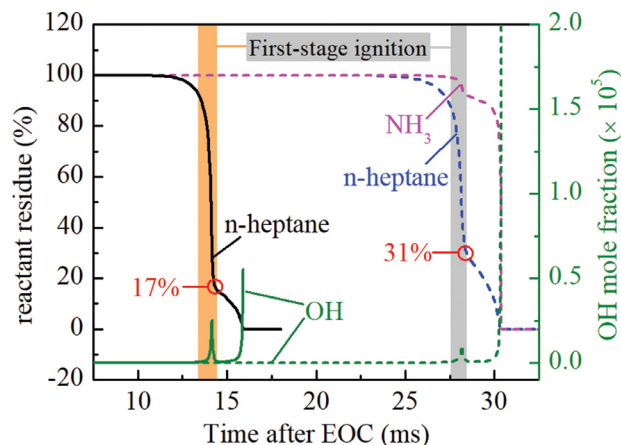


Fig. 8. Simulated reactant consumption and OH radical evolution profiles during mixture 2 autoignition at the initial condition of $\phi = 1.0$, $T_c = 680$ K, and $P_c = 15$ bar. Dashed lines are simulation with NH₃ sub-mechanism and solid lines are simulation without NH₃ sub-mechanism.

the first-stage ignition. About 50% of n-heptane is consumed during the first-stage ignition in contrast to a value of 8% of NH₃. Nevertheless, since the mole fraction of NH₃ is 3.57 times more than that of n-heptane in mixture 2 (see Table 2), 8% NH₃ consumption means a lot of OH radicals are consumed via reaction $\text{NH}_3 + \text{OH} \rightleftharpoons \text{NH}_2 + \text{H}_2\text{O}$. This undoubtedly cuts down the reaction rates of n-heptane low-T oxidation, thus postponing the first-stage ignition. In addition, the remaining n-heptane is almost depleted

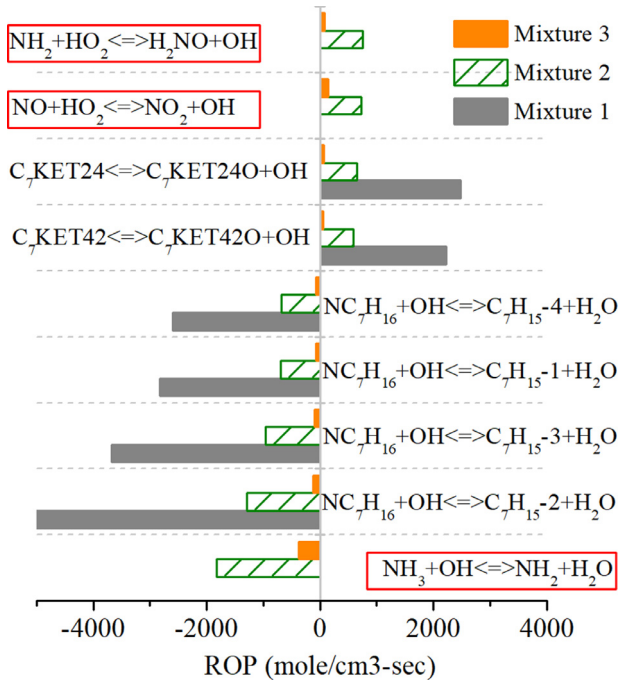


Fig. 9. ROP results of OH radical at the first-stage ignition timepoint of the three blends at the condition of $\varphi = 1.0$, $T_c = 680$ K, and $P_c = 15$ bar. NH₃ related reactions are highlighted by boxes.

in the induction between the first-stage ignition and the final ignition, while there is still 80% NH₃ left prior to the final ignition, which reflects the weak reactivity of NH₃ under the low-T range.

Although NH₃ does not show low-T behavior as n-heptane, it participates in the low-T autoignition of n-heptane/NH₃ blends by passively reacting with free radicals formed in n-heptane oxidation and interacting with carbon-based intermediates. On the other hand, NH₃ acts as a diluent because NH₃ addition reduces the amount of n-heptane in the reactants. Therefore, the effect of NH₃ addition can be divided into two parts, one is to participate in chemical reactions, and the other is to dilute n-heptane. By removing NH₃ sub-mechanism from the mechanism and retaining NH₃'s thermodynamics, another simulation was performed to explore the chemical reaction effect of NH₃ on n-heptane autoigni-

tion. The results are shown in Fig. 8. Firstly, a pronounced phenomenon is that the autoignition timepoint is greatly advanced by halving the first-stage IDT. The result indicates that NH₃ addition affects n-heptane autoignition by acting on its first-stage ignition rather than the second-stage IDT, contrary to the experimental result in Fig. 4. Secondly, compared to the former case (with NH₃ sub-mechanism), more n-heptane was consumed during the first-stage ignition (residue: 17% vs 31%), along with higher OH radical production. According to Section 3, the blending mechanism overpredicts the inhibition effect of NH₃ on n-heptane autoignition at low temperatures in that the first-stage IDTs are significantly overestimated. Combined with the observation in Fig. 8, we have good reasons to speculate that the chemical reaction effect of NH₃ on the first-stage ignition is greatly overestimated by the blending mechanism.

Given that the main deficiency of the current mechanism lies in the prediction on the first-stage ignition, the ROP analysis of OH radical was carried out at the first-stage ignition timepoint of the three blends with the same initial condition as Fig. 8. The ROP results are shown in Fig. 9. For mixture 1, the branching reactions of two ketohydroperoxides species (C₇KET24 and C₇KET42) are two of the most important sources of OH radical which is mostly consumed by H-atom abstraction reactions between OH and NC₇H₁₆, consistent with the well-recognized low-T reaction pathways of n-heptane. For mixtures 2 and 3, it is not surprising that the reaction NH₃ + OH $\rightleftharpoons$ NH₂ + H₂O has the highest negative ROP, accounting for 19.3% OH consumption. However, it is interesting that two NH₃ related propagation reactions NH₂ + HO₂ $\rightleftharpoons$ H₂NO + OH and NO + HO₂ $\rightleftharpoons$ NO₂ + OH exhibit comparable positive ROP to the n-heptane branching reactions, as highlighted in Fig. 9. Song et al. [33] pointed out that the main oxidation pathways of NH₃ at low-to-intermediate temperatures are NH₃ $\rightarrow$ NH₂ $\rightarrow$ H₂NO $\rightarrow$ HNO $\rightarrow$ NO $\rightarrow$ N₂, and the fate of those intermediates are largely determined by HO₂ radical. During the autoignition, HO₂ radical builds up a considerable content, which promotes the oxidation of NO to NO₂ via the active reaction NO + HO₂ $\rightleftharpoons$ NO₂ + OH [15]. In addition, that NH₂ radical transforms into H₂NO radical is dominated by the reactions NH₂ + NO₂ $\rightleftharpoons$ H₂NO + NO and NH₂ + HO₂ $\rightleftharpoons$ H₂NO + OH. Since HO₂ and NH₂ radicals are formed in significant quantities during the autoignition, the reaction NH₂ + HO₂ $\rightleftharpoons$ H₂NO + OH proceeds at an extremely fast speed. Another noteworthy result is

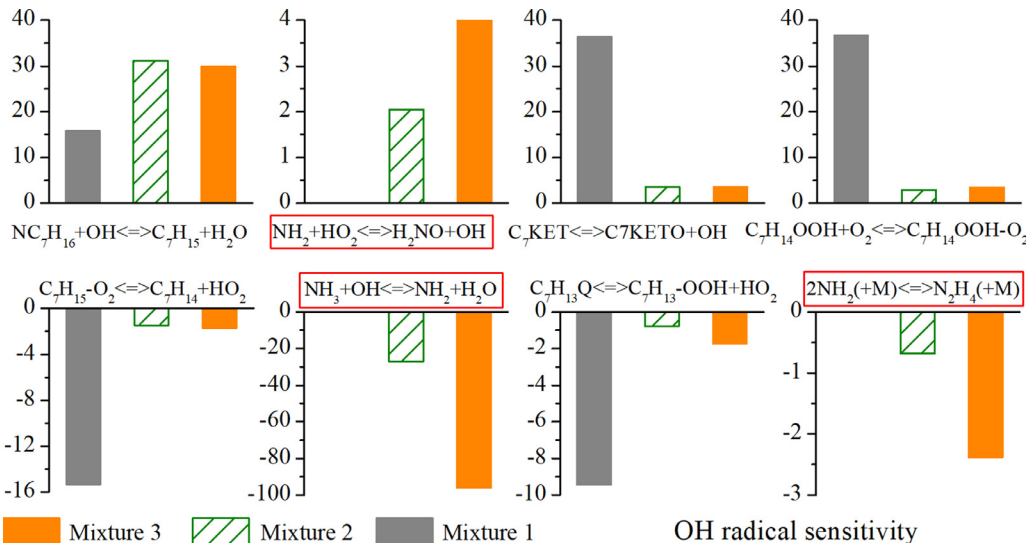


Fig. 10. OH radical sensitivities at the timepoint of first-stage ignition for the three blends at the initial condition of $\varphi = 1.0$, $T_c = 680$ K, and $P_c = 15$ bar. NH₃ related reactions are highlighted by boxes.

that with the increase of NH_3 fraction, the ROP of all reactions, including the reactions from NH_3 sub-mechanism, significantly decreases. This phenomenon reflects the dominant role of n-heptane in OH radical production.

At last OH radical sensitivity analysis was performed to directly locate ignition controlling reactions. The most sensitive twenty reactions are selected and summarized to eight reaction classes for presentation, as shown in Fig. 10. For example, the sensitivity of reaction $\text{C}_7\text{H}_{15}-\text{O}_2 \rightleftharpoons \text{C}_7\text{H}_{14} + \text{HO}_2$ is the average of its six isomer reactions. It should be pointed out that a positive (negative) sensitivity indicates an ignition-promoting (inhibition) effect. There are three NH_3 -related reactions involved whose sensitivities all surge as the increase of NH_3 fraction in the blending fuels. It is found that the reaction $\text{NH}_3 + \text{OH} \rightleftharpoons \text{NH}_2 + \text{H}_2\text{O}$ exhibits the largest negative sensitivity at 20% and 40% NH_3 blends as expected. This reaction consumes a lot of radicals and lowers the rate of the low-T reaction of n-heptane, thus greatly delaying the first-stage ignition. In addition, this reaction is the main pathway through which ammonia acts on n-heptane autoignition. To update the blending mechanism, special attention should be paid to the low-T rate rules of this reaction [15,34]. The only positive sensitivity reaction of the NH_3 subset is the reaction $\text{NH}_2 + \text{HO}_2 \rightleftharpoons \text{H}_2\text{NO} + \text{OH}$. This reaction consumes HO_2 radical and produces active OH radical, promoting the autoignition of NH_3 /n-heptane blend. Important ignition-promoting reactions are all n-heptane related reactions, whose sensitivities significantly decrease with the increase of NH_3 fraction. This is because their promotion effect on the low-T chain branching sequence of n-heptane greatly weakens in the face of the strong OH predatory ability of NH_3 . Reaction $\text{NC}_7\text{H}_{16} + \text{OH} \rightleftharpoons \text{C}_7\text{H}_{15} + \text{H}_2\text{O}$ is an exception because it is itself a OH consuming reaction. Therefore, it will compete for OH radical with reaction $\text{NH}_3 + \text{OH} \rightleftharpoons \text{NH}_2 + \text{H}_2\text{O}$, and has a larger sensitivity when NH_3 is added to n-heptane. Based on the analysis on Figs. 9 and 10, important reactions responsible for the first-stage ignition of n-heptane/ NH_3 blends are identified.

5. Conclusions

This study provides the first low-T autoignition experiment of n-heptane/ NH_3 blends, aiming to reveal the effect of NH_3 addition on n-heptane's low-T autoignition and to provide a preliminary attempt for the development of hydrocarbon/ NH_3 blending mechanism. In this work, IDTs of five n-heptane/ NH_3 blends with NH_3 fractions of 0%, 20%, and 40% were measured in a rapid compression machine for equivalence ratios of 1.0 and 2.0 at pressures of 10 and 15 bar. It is found that both first-stage and total IDTs of the blends become longer with increasing NH_3 fraction at the investigated conditions. Besides, it is observed that their NTC intensity and temperature range of the blending fuels are almost unaffected by NH_3 addition. An interesting phenomenon is that the ratio of first-stage IDT in the total IDT decrease with NH_3 addition, which indicates that NH_3 addition has a greater impact on the second-stage IDT than the first-stage IDT.

An n-heptane/ NH_3 blending mechanism was compiled by merging the n-heptane mechanism of Zhang et al. [31] and the NH_3 mechanism of Glarborg et al. [15]. The blending mechanism can qualitatively capture the changes of IDTs with NH_3 blending ratio, temperature, equivalence ratio and oxygen mole fraction, but significantly overpredicts the IDTs at temperatures below 720 K. Besides, it was further confirmed that the big differences between the experiments and the simulation are mainly from the excessively overpredicted first-stage IDT. Kinetic analyses were conducted on the autoignition process of the blending fuels. Simulation shows that the autoignition of NH_3 /n-heptane blend is initiated by n-heptane low-T oxidation. Besides, the chemical reaction effect of NH_3 greatly prolongs the first-stage IDT of NH_3 /n-heptane

blend. The ROP analysis shows that NH_3 actively participates in the first-stage ignition of the blend both in producing and consuming OH radical. Sensitivity analysis finds that the reactions from the NH_3 subset all surges with increasing NH_3 fraction. In addition, most of the ignition-promoting reactions are related to n-heptane sub-mechanism whose sensitivities decrease with NH_3 addition except the H-atom abstraction reaction of n-heptane with OH radical. Further mechanism development will benefit from those results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors gratefully acknowledge the financial support from National Natural Science Foundation of China (Nos. 51961135105 and 51425602).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.03.019.

References

- [1] R.K. Lyon, D. Benn, Kinetics of the $\text{NO}-\text{NH}_3-\text{O}_2$ reaction, Symp. (Inst.) Combust. 17 (1979) 601–610.
- [2] A. Valera-Medina, H. Xiao, M. Owen-Jones, W.I.F. David, P.J. Bowen, Ammonia for power, Prog. Energy Combust. Sci. 69 (2018) 63–102.
- [3] D.C. Bull, A shock tube study of the oxidation of ammonia, Combust. Flame 12 (1968) 603–610.
- [4] T. Takeyama, H. Miyama, A shock-tube study of the ammonia-oxygen reaction, Symp. (Inst.) Combust. 11 (1967) 845–852.
- [5] L.J. Drummond, High temperature oxidation of ammonia, Combust. Sci. Technol. 5 (1972) 175–182.
- [6] H. Miyama, Ignition of ammonia-oxygen mixtures by shock waves, J. Chem. Phys. 48 (1968) 1421–1422.
- [7] T. Takeyama, H. Miyama, Kinetic studies of ammonia oxidation in shock waves. II. The rate of ammonia consumption, Bull. Chem. Soc. Jpn. 39 (1966) 2352–2355.
- [8] H. Miyama, Kinetic studies of ammonia oxidation in shock waves. IV. Comparison of induction periods for the ignition of $\text{NH}_3-\text{O}_2-\text{O}_2$ with those for NH_3-O_2 -Ar mixtures, Bull. Chem. Soc. Jpn. 41 (1968) 1761–1765.
- [9] N. Fujii, H. Miyama, M. Koshi, T. Asaba, Kinetics of ammonia oxidation in shock waves, Proc. Combust. Inst. 18 (1981) 873–883.
- [10] J.N. Bradley, R.N. Butlin, D. Lewis, Oxidation of ammonia in shock waves, Trans. Faraday Soc. 64 (1968) 71–78.
- [11] B. Shu, S.K. Vallabhuni, X. He, G. Issayev, K. Moshhammer, A. Farooq, et al., A shock tube and modeling study on the autoignition properties of ammonia at intermediate temperatures, Proc. Combust. Inst. 37 (2019) 205–211.
- [12] O. Mathieu, E.L. Petersen, Experimental and modeling study on the high-temperature oxidation of ammonia and related NO_x chemistry, Combust. Flame 162 (2015) 554–570.
- [13] M. Pochet, V. Dias, B. Moreau, F. Foucher, H. Jeanmart, F. Contino, Experimental and numerical study, under LTC conditions, of ammonia ignition delay with and without hydrogen addition, Proc. Combust. Inst. 37 (2019) 621–629.
- [14] X. He, B. Shu, D. Nascimento, K. Moshhammer, M. Costa, R.X. Fernandes, Auto-ignition kinetics of ammonia and ammonia/hydrogen mixtures at intermediate temperatures and high pressures, Combust. Flame 206 (2019) 189–200.
- [15] P. Glarborg, J.A. Miller, B. Ruscic, S.J. Klippenstein, Modeling nitrogen chemistry in combustion, Prog. Energy Combust. Sci. 67 (2018) 31–68.
- [16] J.A. Miller, M.D. Smooke, R.M. Green, R.J. Kee, Kinetic modeling of the oxidation of ammonia in flames, Combust. Sci. Technol. 34 (1983) 149–176.
- [17] R.P. Lindstedt, F.C. Lockwood, M.A. Selim, Detailed kinetic modelling of chemistry and temperature effects on ammonia oxidation, Combust. Sci. Technol. 99 (1994) 253–276.
- [18] Z. Tian, Y. Li, L. Zhang, P. Glarborg, F. Qi, An experimental and kinetic modeling study of premixed $\text{NH}_3/\text{CH}_4/\text{O}_2/\text{Ar}$ flames at low pressure, Combust. Flame 156 (2009) 1413–1426.
- [19] T. Mendiara, P. Glarborg, Ammonia chemistry in oxy-fuel combustion of methane, Combust. Flame 156 (2009) 1937–1949.
- [20] H. Nakamura, S. Hasegawa, T. Tezuka, Kinetic modeling of ammonia/air weak flames in a micro flow reactor with a controlled temperature profile, Combust. Flame 185 (2017) 16–27.

- [21] K.P. Shrestha, L. Seidel, T. Zeuch, F. Mauss, Detailed kinetic mechanism for the oxidation of ammonia including the formation and reduction of nitrogen oxides, *Energy Fuels* 32 (2018) 10202–10217.
- [22] B. Mei, X. Zhang, S. Ma, M. Cui, H. Guo, Z. Cao, et al., Experimental and kinetic modeling investigation on the laminar flame propagation of ammonia under oxygen enrichment and elevated pressure conditions, *Combust. Flame* 210 (2019) 236–246.
- [23] X. Han, Z. Wang, M. Costa, Z. Sun, Y. He, K. Cen, Experimental and kinetic modeling study of laminar burning velocities of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{CH}_4/\text{air}$ premixed flames, *Combust. Flame* 206 (2019) 214–226.
- [24] K. Takizawa, A. Takahashi, K. Tokuhashi, S. Kondo, A. Sekiya, Burning velocity measurements of nitrogen-containing compounds, *J. Hazard. Mater.* 155 (2008) 144–152.
- [25] A.J. Reiter, S. Kong, Combustion and emissions characteristics of compression-ignition engine using dual ammonia-diesel fuel, *Fuel* 90 (2011) 87–97.
- [26] L. Yu, Y. Qiu, Y. Mao, S. Wang, C. Ruan, W. Tao, et al., A study on the low-to-intermediate temperature ignition delays of long chain branched paraffin: iso-cetane, *Proc. Combust. Inst.* 37 (2019) 631–638.
- [27] L. Yu, Y. Mao, Y. Qiu, S. Wang, H. Li, W. Tao, et al., Experimental and modeling study of the autoignition characteristics of commercial diesel under engine-relevant conditions, *Proc. Combust. Inst.* 37 (2019) 4805–4812.
- [28] L. Yu, Y. Mao, A. Li, S. Wang, Y. Qiu, Y. Qian, et al., Experimental and modeling validation of a large diesel surrogate: autoignition in heated rapid compression machine and oxidation in flow reactor, *Combust. Flame* 202 (2019) 195–207.
- [29] H. Hu, J. Keck, Autoignition of adiabatically compressed combustible gas mixtures, *SAE Technical Paper* (1987): 872110.
- [30] C.-J. Sung, H.J. Curran, Using rapid compression machines for chemical kinetics studies, *Prog. Energy Combust. Sci.* 44 (2014) 1–18.
- [31] K. Zhang, C. Banyon, J. Bugler, H.J. Curran, A. Rodriguez, O. Herbinet, et al., An updated experimental and kinetic modeling study of n-heptane oxidation, *Combust. Flame* 172 (2016) 116–135.
- [32] CHEMKIN-PRO 15131, Reaction Design, San Diego, 2013.
- [33] Y. Song, H. Hashemi, J.M. Christensen, C. Zou, P. Marshall, P. Glarborg, Ammonia oxidation at high pressure and intermediate temperatures, *Fuel* 181 (2016) 358–365.
- [34] S. Salimian, R.K. Hanson, C.H. Kruger, High temperature study of the reactions of O and OH with NH_3 , *Int. J. Chem. Kinet.* 16 (1984) 725–739.